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REVIEW ARTICLE

Exploring the modern approaches to enhance fungal endophyte-derived bioactive secondary metabolites

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Abstract

Over the past few decades, microbial-derived bioactive compounds have been tested for antiviral, antimicrobial, and anticancer properties. In addition, fungal-derived bioactive secondary metabolites (SMs) are increasingly being suggested as suitable alternative sources of potent bioactive compounds. The development of suitable, precise in vitro and in vivo screening techniques may contribute to identifying the biochemical and physiological effects of compounds. This advancement in bioassay evaluation techniques helps identify potential bioactive microbes rapidly. The main obstacles, however, have been the production of insufficient amounts of chemicals, endophytes' attenuation or loss of ability to produce the molecule of interest when grown in cultures, and fungal endophytes' failure to exhibit their full biosynthetic potential in lab conditions. These have led to the use of small chemical elicitors that activate the silent biosynthetic gene clusters (BGCs) in fungi, causing epigenetic alterations that increase the amount of desired metabolites or trigger the synthesis of hitherto unknown compounds. The silent BGCs were activated to maximize production of bioactive secondary metabolites, thereby increasing the yield of desired compounds or triggering the synthesis of novel metabolites. Other strategies include gene knocking, inducing mutations, heterologous expression, one strain-many compounds (OSMAC), epigenetic modifications, etc. This review is focused on the mechanism of plant–microbe interaction in enhancing the biosynthesis of fungal metabolites along with the BGCs for the biosynthesis of the bioactive fungal metabolites. Furthermore, we also discuss the genomic mining approaches for BGCs, the role of ribosomal engineering, precursor feeding, and various elicitors to explore the structural diversity of novel bioactive compounds.

Keywords Co-culturing · Endophytic fungi · Epigenetic modification · G*enomic mining* · OSMAC · Precursor feeding · Ribosomal engineering

Introduction

Some plants possess a life cycle of two or more years (called perennials) and are usually infected by some microorganisms that form colonies inside the cells and tissues of these plants and utilize this space for the completion of their life cycles. These microbes are termed endophytic microorganisms. However, they do not necessarily cause any apparent harm to the plant or affect the morphology of their host cells. Some of them might also be beneficial. Fungi and bacteria are the two major endophytic microbes that are found within plant cells and tissues, and they form a relationship that benefits both microbes as well as plants, i.e., symbiosis (Kandel et al. 2017; Singh et al. 2024).

The endophytic microorganisms can be classified into two major categories: facultative microbes and obligate

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microbes. These microbes are able to form colonies of different kinds. Obligate microbes require strict aerobic or anaerobic conditions for their survival, whereas facultative microbes have their specific survival conditions (aerobic/anaerobic) but can also survive in the absence of those conditions (Kaur et al. 2023). The endophytic microorganisms that are mostly facultative in nature tend to infect the plant cells and tissues during specific phases of their life cycles and are found on the exterior of the plant cell surface such that they can form an organized relationship (might be symbiotic) with the host plant root and soil (rhizosphere soil). In contrast, the obligate endophytes tend to remain inside the host plant cells for their entire life period. These endophytes usually spread and multiply among the plants from seeds to seedlings and hence to newer plant generations (i.e., vertical transmission) and utilize or change the processes involved in their host cell's metabolism in such a way that renders certain products that are essential for the survival and continuity of such pathogens (Dwibedi and Saxena 2018). Amidst diverse and distinct varieties of endophytes, the endophytic fungi have drawn much more exploration interest from researchers. This is not only due to the reservoir of unique compounds with anticarcinogenic and bactericidal properties but also because they provide some biological precursors and catalysts that help in the production of essential bio-oils (Dwibedi et al. 2019). Basically, the endophytic fungi are characterized as mitosporic and meiosporic ascomycetes found in the internal tissues of the host plant. The host plant secretes certain defensive compounds to prevent the endophyte's colonization; in response to these defensive compounds, the endophytes secrete detoxifying enzymes (cellulases, lactases, xylanases, and proteases). These protective enzymatic actions of endophytes enable them to colonize and reside inside the host plant as dormant or hidden throughout the entire lifespan. The host plants also show symbiotic relationships with fungal endophytes, which benefit the plant with disease resistance, help in plant growth, and stimulate the production of SMs. Endophytic fungal diversity varies with factors, such as host plant genotype and physiology, environmental conditions, and anthropogenic influences. For example, fungal endophytic diversity has been found to be much higher in the tropics and subtropics compared to other regions (Dwibedi and Saxena 2018; Dwibedi and Sazexa 2019; Saini et al. 2024). Endophytic fungi produce some vital SM compounds that help plants to survive under biotic and abiotic stresses, such as temperature, ultraviolet radiation, pathogens, viruses, pH, and heavy metals (Basit et al. 2024). Fungi are conserved as reservoirs of bioactive SMs and precursors for drug development and formulations. The SMs belong to a diverse family of organic molecules, such as alkaloids, phenolics, terpenoids, peptides, and steroids, and can broaden their important biological applications for pharmacological or medical purposes (Jamtsho et al. 2024). In 1898, Vogle isolated the first endophytic fungus from Perennial ryegrass (*Loliumtum eletum*) seeds (Zheng et al. 2021). Some of the SMs synthesized from these fungal endophytes have a lower molecular weight and are not actively involved in primary cellular processes such as cell growth and development or reproduction. They are synthesized only when the cellular division and enlargement process stops (Bind et al. 2022). In addition to being so favorable, one of the major challenges is getting a higher yield of these biologically active and useful chemical compounds. This prompts the use of epigenetics as a vital system that can change the genetic expressions encoding for the SMs, in order to retrieve higher amounts of the desired products from these endophytes (Gupta et al. 2020). The study of the process that involves any change that can be inherited from one generation to another but does not include any alteration in the DNA sequences is called epigenetics (Al Aboud et al. 2022). These epigenetic changes, also referred as epigenetic modifications, modify, and redesign the chromatin content through various histone protein changes, i.e., post-translational modifications, RNA interference, DNA methylation, etc. A bunch of genes, called BGCs, code for SMs and occur in an inactive state of chromatin (heterochromatin) where the active and continuously expressed constitutive genes are transcribed via epigenetics. Thus, tiny molecules that can be used as epigenetic modifiers can alter the chromatin framework and govern the processes through which silent and non-constitutive genes can be activated. Henceforth, they might be reasonably utilized for getting these novel and biologically active chemical compounds. These natural compounds have a high utility in therapeutics. To maximize their production, the silent BGCs were activated through either a change in the conditions required to culture microorganisms or cultivating two or more microbes together, i.e., co-cultivation (Bind et al. 2022). Gene knocking, inducing mutations, co-cultivation/co-culturing, heterologous expression, OSMAC, and epigenetic modification are some of the major strategies used for this purpose (Deshmukh et al. 2022).

In this review, the interactions between plants and endophytic fungi, along with the genetic makeup of SMs, are discussed. Further, the various cultivation-dependent approaches, such as the OSMAC approach, co-cultivation approaches, and genome mining, are thoroughly included. The review also tends to recapitulate the activator–precursor feeding (in ribosomal engineering) and focuses on the epigenetic modulation approach in endophytic fungi for an increased production of bioactive molecules.

Phenomenon of interaction between host plant and endophytic fungi

The endophytic fungi usually belong to the class of *Ascomycetes* and are sometimes also found in the classes of *Zygomycetes* and Basidiomycetes (Rungjindamai and Jones 2024). The fungal endophytes are divided into two main groups, namely *Balansiaceous* and non-*Balansiaceous*, which can be further subdivided into four distinct classes as mentioned below. *Balansiaceous* endophytes of grasses were first described in the late nineteenth century on seeds of various *Lolium* species (Dighton and White 2005). Class I contains fungal endophytes that belong to the *Clavicipitaceae* family, including many species, but are restricted to some grasses. However, they can easily transmit vertically to the host plant by seed infection. Class I fungal endophytes are predominantly drought-tolerant and increase the host plant tolerance. They also produce some chemicals that are toxic and provide protection against herbivores. Class II endophytes are the mycorrhizal fungi that help in colonizing various parts of plants, including seed coats, but they do not form intracellular mycorrhizal structures. Class II endophytes can provide stress tolerance to the host plants. Class III endophytes differ from other classes in terms of horizontal propagation and existence. This class of endophytes is extensively found in plant communities that colonize vascular, non-vascular, woody, and herbaceous angiosperms. A single plant can colonize hundreds of endophytes. Class IV endophytes are commonly found in a variety of host plants and diverse habitats ranging from Arctic, Antarctic, alpine, subalpine, and tropical to temperate ecosystems (Hazra et al. 2024).

The possibility of diverse phytochemical synthesis due to plant–endophyte interaction is scientifically important since in vitro synthesis is sometimes highly expensive and unstable. The production of SMs mostly occurs by the activation of biosynthetic pathways connected with plant defense for survival. In this context, fungi act as biological triggers to initiate metabolite biosynthesis in response to plants. Endophytic fungal interactions can be considered balanced interactions with the plant as a host, which requires the activation of a virulence system to escape host defense mechanisms and endophytic fungal colonization. There is a balance to this interaction because the fungus survives on nutrients from the host plant, and, in return, provides benefits such as resistance to biotic and abiotic stress (Baron and Rigobelo 2022). The plant defense system can be induced by the chitin, which is present on the fungal cell wall (Fig. 1) (Gong et al. 2020). To escape from this, the endophytic fungus can *produce* chitin deacetylases (*CDA*) as seen in case of

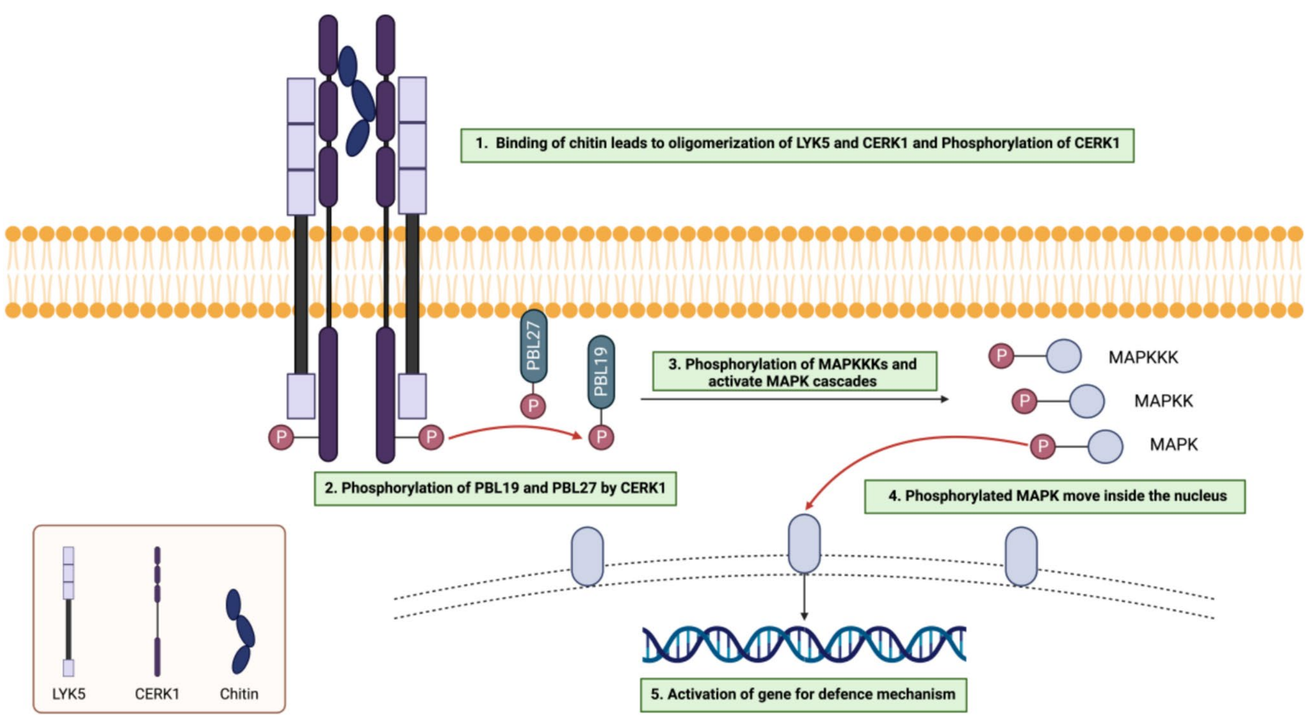

Fig. 1 Fungal–chitin-induced plant defense mechanism, where chitin binding onto the receptors further leads to the activation of oligomers CERK1, which in turn phosphorylate to activate the PBL19 and PBL27, which further activate the MAPK cascade and lead to the activation of genes responsible for the defense mechanism (Gong et al. 2020)

Pestalotiopsis sp., which is an endophytic fungus. The CDA converts chitin into chitosan, helps in evading plant defense mechanisms, and helps in the colonization of endophytic fungi within the host plant (Cord-Landwehr et al. 2016) (Fig. 2).

The interactions between the endophytic fungus and host plant show a variation from the mutualistic relationship to antagonism with the influence of physiological conditions. Factors affecting the endophytic microbes and plant relationship include the nutrient availability for endophytes, metabolic state of the host plant, plant genotype, plant tissue type, and strain type of endophytes. Therefore, the underlying mechanisms of plant–endophyte interactions may influence plant physiology through their modes of response to biotic and abiotic factors. Endophytes use different strategies to influence plant growth and health. Primarily, they produce important SMs, such as antibiotics, toxins, volatile organic compounds, plant hormones, phytoalexins, and reactive oxygen species, which affect plant growth and development. The excessive root growth shown by the plant inoculated with endophytes suggests the stimulation of plant growth by the inoculant endophytes. Several reports demonstrated the synthesis of phytohormones by the endophytic fungi is one of the reasons for promoting plant growth, for instance, phytohormones, such as gibberellic acid produced by *Penicillium commune* and indole acetic acid by *Galactomyces formosus* (Zhang et al. 2019; Sena et al. 2024; Zhang et al. 2024a, b). The production of phytohormones has been reported by strains of *Piriformospora indica* that colonize the plant roots. The production of auxin was found to be stimulated by the *P. indica* in the host by interfering with the auxin biosynthesis and cell signaling of the host and inducing root growth. In addition, some endophytes manipulate the plant host metabolism to alter the uptake of nutrients from the rhizosphere (Kundu et al. 2021).

Epigenetic modification and genetic makeup of fungal secondary metabolites

The plant endophytic microbes secrete SMs, which stimulate the expression of cryptic plant genes without causing any mutation or permanent change. The expression of plant cryptic genes can lead to an enhanced metabolism and growth of plants. Most plants readily recognize biomolecules synthesized from endophytes through their chemo-perception systems. The induction of the plant defense system in response to endophytic fungi response results in

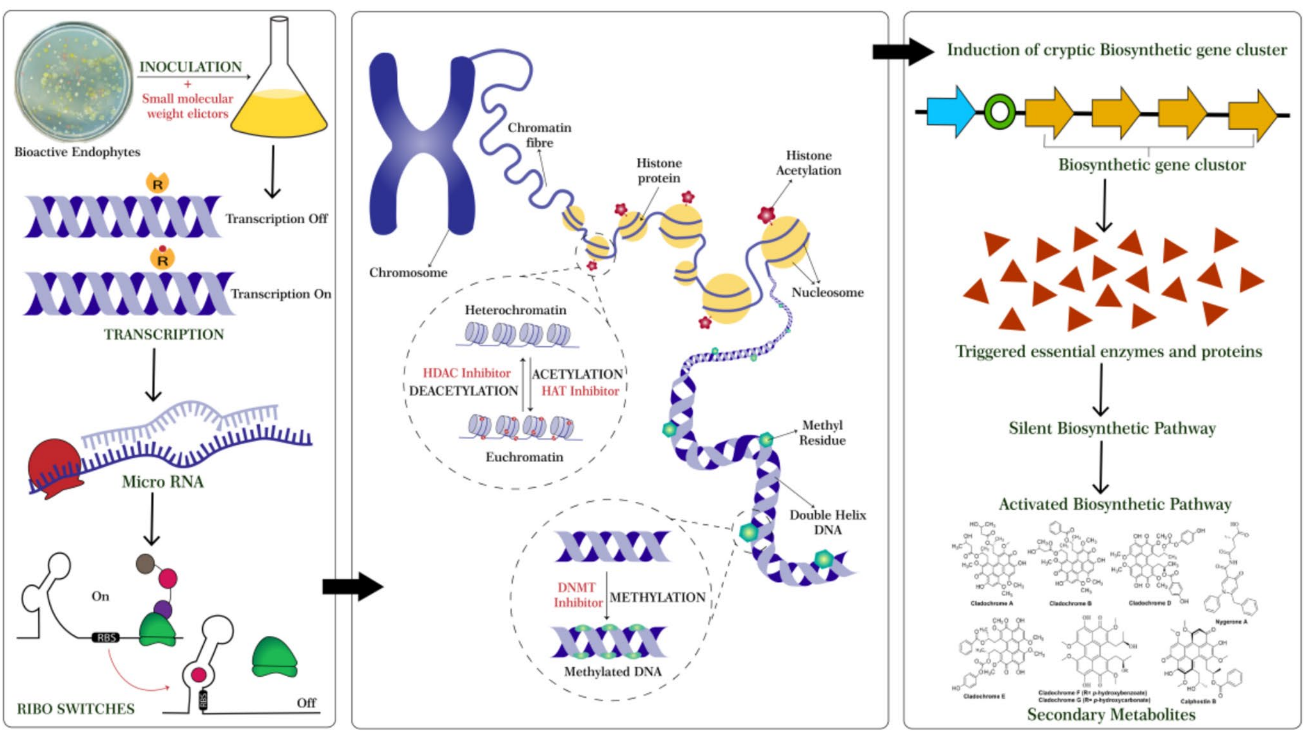

Fig. 2 Epigenetic activation of secondary metabolite production. Inoculation with small molecular weight elicitors activates transcription in bioactive endophytes, regulated by riboswitches and microRNAs (left panel). Chromatin remodeling, mediated by histone modifications and DNA methylation, enables gene accessibility through HDAC, HAT, and DNMT inhibitors (center panel). Activation of silent biosynthetic gene clusters leads to essential enzyme production and initiates a biosynthetic pathway, resulting in secondary metabolites, such as Cladochromes, Nygerone A, and Calphostin B (right panel)

changes in plant metabolic function. Endophytic microbes, alone or in combination, have been shown to enhance the resistance of *Withania somnifera* plants to the expression of virulence genes against biotic stresses by stimulating systemic defense responses (Adeleke and Babalola 2021).

The natural bioactive metabolites produced by fungi can be used in pharmaceutics for drug development and formulation. The biosynthesis of these SMs involves multiple enzymes encoded by multiple genes, which usually appear in the form of BGCs, and these genes are responsible for the biosynthesis of specific SMs (Pfannenstiel and Keller 2019). The SMs can be polyketides, non-ribosomal peptides, and terpenes (Fig. 3). Recent studies showed the presence of polyketide compounds comprising chromones and α-pyrone in endophytic fungi. The plant and endophytic fungi produced the chromones, which showed potent antimicrobial activity against pathogens; for instance, the (*S*)-5-hydroxyl-2-(1-hydroxyethyl)-7-methylchromoneisolated from the endophytic fungi *Bipolaris eleusines* showed potent inhibitory action against S*taphylococcus aureus* (He et al. 2019). Chromones isolated from an endophytic fungus and host plant can be used to develop medicine against pathogenic microbes and are considered an important resource. Chaetoquadrin, D-2,6-dimethyl-5-methoxyl-7-hydroxylchromone, 6-methoxymethyleugenin, isoeugenitol, and 6-hydroxymethyleugenin isolated from *Xylomelasma* sp., an endophytic fungus, showed antimicrobial activity against the pathogens *Agrobacterium tumefaciens*, *Bacillus subtilis*, *Erwinia*

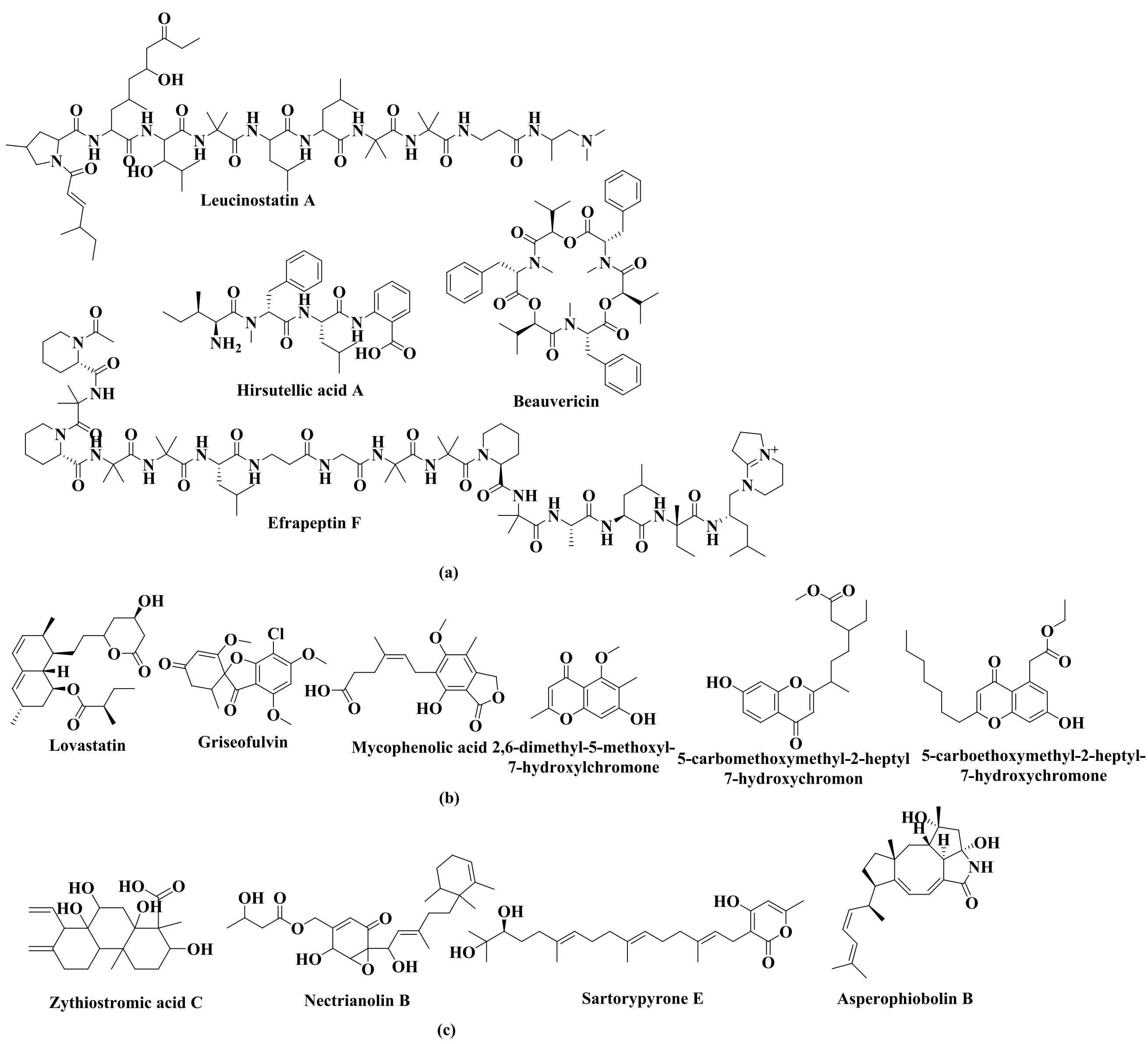

Fig. 3 Natural bioactive metabolites produced by fungi: **a** non-ribosomal peptides; **b** polyketides; and **c** terpenes

carotovora, *Mycobacterium tuberculosis*, *Ralstonia solanacearum*, *Staphylococcus haemolyticus*, *and Xanthomonas vesicatoria*. Chromone and isocoumarin derivatives from the endophytic fungus *Xylomelasma* sp. Samif07 have antibacterial and antioxidant activities (Lai et al. 2021). Polyketides are synthesized by polyketide synthases (PKSs) using an acyl-Coenzyme A (CoA) precursor. PKSs are multi-domain enzymes or enzyme complexes comprising acyltransferase (AT), ketosynthase (KS), thioesterase (TE), and optional domains (Wang et al. 2020). *Trichoderma koningiopsis* QA-3, an endophytic fungus of *Artemisia argyi,* was found to synthesize phytochemicals, which showed antimicrobial activity against human pathogenic *E. coli* (Shi et al. 2017) (Fig. 4).

The BGCs contain certain transcription factors that are highly specific for them, and these factors control all the present genes through a positive regulation mechanism, even the genes that produce proteins that minimize the toxicity of the bioactive molecules yielded from the BGCs themselves (Keller 2019). It is observed that most BGCs are silent rather than active in research lab settings and are present in the heterochromatic part of the genome, because of which it is quite difficult to obtain a higher yield of these compounds. As such many different techniques and procedures have been applied that successfully provide a route into these heterochromatin regions so that they can be transformed into euchromatin regions for activating the BGCs. One such approach is using stimulants for epigenetic regulation and genetically manipulating the proteins responsible for histone modification (Pillay et al. 2022) (Fig. 2).

BGCs are nothing but basically a cluster of well-packed genes that encode for several enzymes that are responsible for the production of the secondary metabolic compounds. For any genome that has been completely sequenced, the number of BGCs can be predicted using various algorithms since the BGCs clusters are arranged with synthases (such as polyketide synthases) and synthetases (non-ribosomal peptide synthetases) that are highly conserved and sequenced in a coupled manner. From about 19 different species of *Aspergillus*, it was observed that they possess at least 21–66 BGCs in their genome per species; this has motivated the researchers to develop certain procedures and strategies to activate these BGCs and render different metabolites with a high utility (Pfannenstiel and Keller 2019). The SMs are usually produced from the primary metabolite groups through various metabolic pathways, and acyl-CoA enzymes are one of the fundamental blocks that build up the various metabolites, such as terpenes

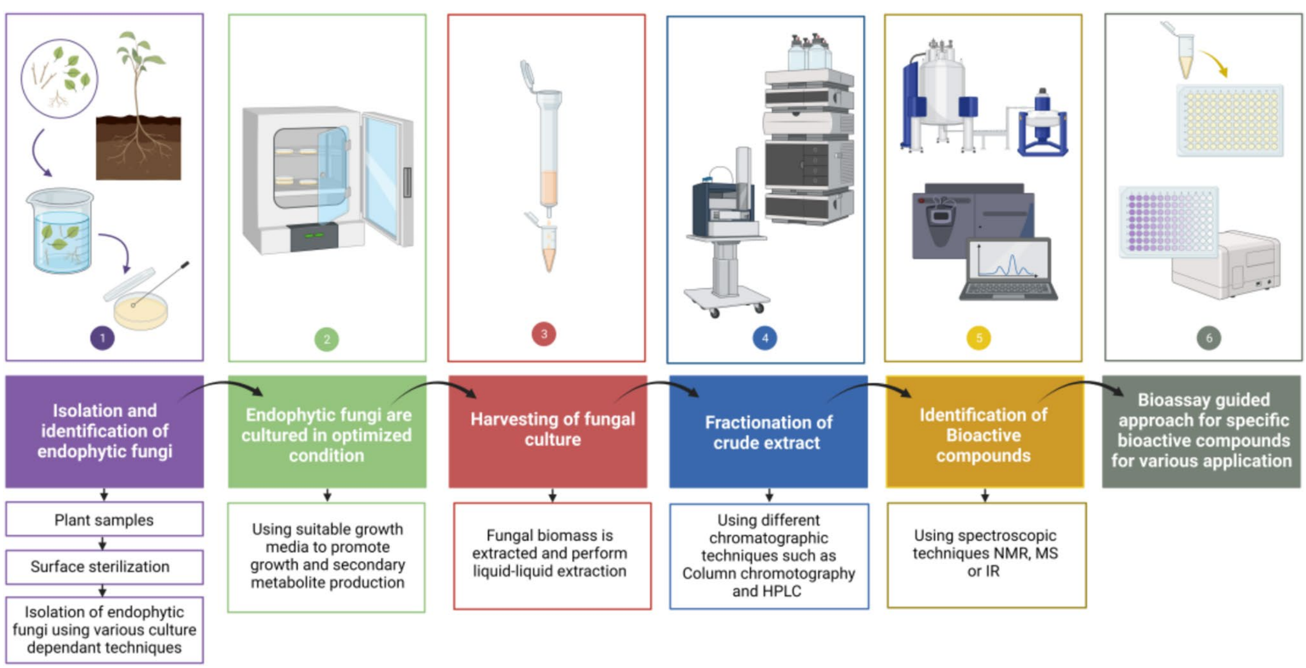

Fig. 4 Steps involved in the cultivation-dependent approach for isolation of bioactive compounds from endophytic fungi. (Step 1—isolation and identification of endophytic fungi from plant samples, Step 2—The isolated fungi are then cultured under optimized conditions using suitable growth media to promote fungal growth and secondary metabolite production, Step 3—Harvesting of fungal cultures involves extracting fungal biomass followed by liquid–liquid extraction, Step 4—The crude extract is fractionated using chromatographic techniques, such as column chromatography and high-performance liquid chromatography (HPLC), to isolate different components, Step 5—Identification of bioactive compounds is performed using advanced spectroscopic techniques like nuclear magnetic resonance (NMR), mass spectrometry (MS), or infrared spectroscopy (IR), Step 6—bioassay-guided approach is employed to identify specific bioactive compounds for various applications)

(carotene), polyketides (aflatoxin), etc. The BGC genes are moreover regulated together in a collateral manner or in sync depending upon the function that is fated for that metabolite. One such case can be seen in *Aspergillus fumigatus* where, during the process of formation of spores, the BGC gets activated, encoding for pigmentation. It is observed that when a plant gets infected with a particular virus and small colony formation begins, then the BGCs present in *Fusarium graminearum* get activated and more regulated to release the virulence factor—trichothecene. Moreover, the *Fusarium* sp. produces bikaverin, which is antibacterial in nature and gets expressed only when there is a conflict due to the presence of another bacteria, namely *Ralstonia solanacearum* (Keller 2019) (Fig. 2).

The number of genes present in BGCs varies from two in valactamide to more than twenty in Aflatoxin. The BGCs that have 2–4 genes are responsible for encoding synthase and synthetase enzymes and additional enzymes that modify the very first enzymes of these metabolites. In addition to having the genes for encoding synthases and synthetases and a few other enzymes, the large BGCs also possess genes contributing to specific transcription factors of respective clusters (Mózsik et al. 2022). The majority of the SMs are produced by such genes in proximity, but there are also some exceptions. It was seen in *Aspergillus fumigatus* that there are some genes in the so-called “supercluster” that encode for two different products, fumagillin and pseurotin, yet are interlaced at the same location in the genome. Nidulanin A can only be synthesized when the enzymes present on two different chromosomes act together in *Aspergillus nidulans* while the trichothecene secreted by certain *Fusarium* sp. has its major encoding gene in one primary cluster—the tri5 BGC—and the remaining three are present on the exterior of this cluster. A toxin obtained from plants, *Dothiostromin*, has a somewhat similar structure to that of Aflatoxin and is obtained from *Dothistroma septosporum*, such that the genes responsible for its production are split into three small clusters but are found on the same genomic chromosome (Keller 2019). The BGCs in some endophytic fungi are usually present adjacent to those of the telomeric region in the heterochromatin genome. These biosynthetic genes are regulated via epigenetic modification involving processes, such as methylation of DNA and deacetylation of histones. The environmental conditions, such as biotic and abiotic stresses, also promote the chromatin-dependent activation of the silent BGCs to produce the metabolites. The nutrient availability and developmental stage of fungi also play an essential role in triggering the induction of clusters. Although secondary metabolism might seem to have a lesser importance in the growth and development processes of an organism, the secondary products affect the survival of the organism in its surroundings and thus regulation of BGCs is necessary (Pillay et al. 2022) (Table 1).

Chromatin and its role in the control of gene expression

The modification of the chromatin material is necessary for normal cell function and to administer the variations in the transcription. These alterations and variations get regulated by some factors that might be intrinsic (inside the cell) or extrinsic (external ecological conditions) and further impact the maintenance of the cellular structure, growth, differentiation, catabolic and anabolic activities, etc. (Cheung and Lau 2005; Khaldi et al. 2010; Giaimo et al. 2019). The two major enzymes, i.e., histone deacetylases (HDACs) and histone acetyltransferases (HATs), regulate these internal factors and modify the chromatin material via a local modification system involving the elimination, establishment, and preservation of chromatin, deciding upon whether to induce the transcription process or not. Whether or not a typical secondary compound produced by BGCs will be expressed or repressed in a fungal organism is usually determined by the histone-binding proteins, histone acetyltransferase, and methyl-transferases. Hence, they are also termed epigenetic modifiers. Two of these HDAC inhibitors and DNA methyl-transferase (DNMT) inhibitors help in the production of metabolites since they act as elicitors. The acetyltransferases and deacetylases are quite hostile and have an adverse effect on the acetylation of histones. Acetyltransferases enable the transcription process, whereas deacetylases are linked to the silencing of genes. Hence, if these HDACs are deactivated, removed, or inhibited via some chemicals (certain elicitors), then the yield of metabolites produced by filamentous fungi can be enhanced (Pillay et al. 2022).

Research on the 3D structure of chromosomes has shown that the spatial arrangement of gene clusters plays a significant role in determining their activities. Silent and actively expressed gene clusters exhibit distinct spatial configurations, with actively expressed clusters typically being located in more accessible regions of chromatin. This spatial arrangement enables efficient transcription by exposing promoter regions to transcription factors, while silenced clusters remain tightly packed and inaccessible. In their study, (Collemare and Seidl 2019) have demonstrated that the metabolic pathway gene clusters can form independent 3D domains within chromatin, allowing coordinated regulation of genes involved in secondary metabolite production. Technologies like assay for transposase accessible chromatin sequencing (ATAC-seq) have been instrumental in analyzing chromatin openness and identifying regions with accessible chromatin where active gene clusters reside. ATAC-seq is a powerful tool for mapping accessible chromatin regions and has provided insights into how chromatin remodeling and spatial configuration

Table 1 A list of novel bioactive compounds reported from applying various epigenetic modifications, along with known biological activity

Sr. No	Endophytic Fungi	Culture Conditions	Epigenetic Approach	Novel Bioactive Compounds	Biological Activity	References
Co-culturing approach to bioprospecting of novel bioactive compounds' isolation/production						
1	*Aspergillus* sp. and *Aspergillus* sp.	GYP medium, 30 °C, 5–7 days	Co-cultivation	Aspergicin, Neoaspergillic acid, and Ergosterol	Anti-bacterial activity against selected Gram bacteria	Zhu et al. (2011)
2	*Unidentified fungus (E33)* and *Unidentified fungus (K38)*	GYP medium, 30 °C, 5–7 days	Co-cultivation	8-hydroxy-3-methyl-9-oxo-9H-xanthene-1-carboxylic acid methyl ether	Antifungal activity	Li et al. (2017)
3	*Pestalotia* sp. and *unidentified bacterium*	GYP medium, 25 °C, 6 days, 250 rpm	Co-cultivation	Pestalone	Antibiotic activity against MRSA and VREF	Marmann et al. (2014)
4	*Chaunopycnis* sp. and *Trichoderma hamatum*	ISP2 medium, 26.5 °C, 5 days, 180 rpm	Co-cultivation	Chaunopyran A, Methyl-pyridoxatin	Antimicrobial and antifungal activity	Peng et al. (2021)
5	*Aspergillus sulphureus* + *Isaria felina*	ISP2 medium, 25 °C, 5 days, 180 rpm	Co-cultivation	17-hydroxynotoamide D, 17-O-ethylnotoamide M, 10-O-acetylsclerotiamide, 10-O-ethylsclerotiamide, 10-O-ethylnotoamide R	Anticancer activity	Peng et al. (2021)
6	*Phoma* sp. and *Armillaria* sp.	PDB medium, 28 °C, 7 days, 180 rpm	Co-cultivation	Phexandiols A and B, phomesters A–C	Antimicrobial and cytotoxic activity	Peng et al. (2021)
7	*Trichoderma harzianum and Talaromyces pinophilus*	PDB medium, 28 °C, 21 days, static condition	Co-cultivation	Harziaphilic acid	Anticancer activity	Vinale et al. (2017)
8	*Talaromyces aculeatus and Penicillium variabile*	Liquid media, 28 °C, 9 days, 180 rpm	Co-cultivation	Penitalarins A–C, Nafuredin A, and Nafuredin B	Anticancer activity	Zhuang and Zhang (2021)
9	*Penicillium crustosum and Xylaria* sp.	Solid medium, 28 °C, 30 days, static condition	Co-cultivation	Penixylarins A–D	Antituberculosis	Peng et al. (2021)
10	*Chaetomium* sp. *and Pseudomonas aeruginosa*	Solid medium, 28 °C, 35 days, static condition	Co-cultivation	Chaetoisochorismin, Shikimeran B, Chaetobutenolides A-C, WF-3681 methyl ester	Antimicrobial and cytotox activity	Peng et al. (2021)
11	*Aspergillus fumigatus and Streptomyces leeuwenhoekii*	GYE medium, 30 °C, 3 days, 180 rpm	Co-cultivation	Luteoride D, Pseurotin G	NA	Peng et al. (2021)
12	*Aspergillus versicolor* and *Bacillus subtilis*	Solid medium, 23 °C, 28 days, static condition	Co-cultivation	Aspvanicins A, B	Cytotoxic activity and antibacterial activity	Peng et al. (2021)
13	*Rhinocladiella similis* and *Streptomyces rochei*	ISP2 medium, 28 °C, 25 days, 180 rpm	Co-cultivation	Borrelidins J and K, 7-methoxy-2,3-dimethylchromone-4-one, Borrelidin,	Antibiotic activity against MRSA	Yu et al. (2019)
14	*Aspergillus ochraceus* and *Bacillus subtilis*	Solid medium, 40 °C, 15 days, static condition	Co-cultivation	Waspergillamide B, Ochraspergillic acids A and B	Cytotoxic activity	Frank et al. (2019)

Table 1 (continued)

Sr. No	Endophytic Fungi	Culture Conditions	Epigenetic Approach	Novel Bioactive Compounds	Biological Activity	References
15	*Penicillium citrinum* and *Pantoea agglomerans*	ISP2 medium, 28 °C, 8 days, 180 rpm	Co-cultivation	Pulicatin H, Pulicatin F	Antifungal activity	Thissera et al. (2020)
Epigenetic modifier approach to bioprospecting of novel bioactive compounds' isolation/production						
16	*Chaetomium globoporum*	Liquid media, 28 °C, 7 days, 160 rpm	5-azacytidine	Globosporins A and B, Globosporins C and D	Antibiotic activity	Cai et al. (2022)
17	*Arcopilus aureus*	PDB medium, 26 °C, 7 days, 120 rpm	5-azacytidineandSAHA	Resveratrol	Antistaphylococcal and anti-Candida activity	Dwibedi and Saxena (2022)
18	*Aplosporella javeedii*	Solid medium, 28 °C, 20 days, static condition	OSMAC	Aplosporellins A–K, Pramanicin A	Anticancer activity	Gao et al. (2020)
19	*Phomopsis liquidambaris*	PDB medium, 28 °C, 7 days, 180 rpm	Precursor feeding	Naringenin, Kaempferol	NA	Yang et al. (2021)
20	*Hypoxylon anthochroum*	PDB medium, 28 °C, 21 days, 180 rpm	DNMT and HDAC inhibitors	NA	Antioxidant and antimicrobial activity	Mishra et al. (2020)
21	*Diaporthe* sp.	PDB medium, 30 °C, 30 days, 180 rpm	SAHA	Piperine	NA	Jasim et al. (2019)
22	*Chaetomium globosum and Paraconiothyrium brasiliense*	Liquid media, 25 °C, 6 days, 110 rpm	Fungal elicitor	Paclitaxel	Anticancer activity	Salehi et al. (2019)
23	*Pestalotiopsis microspore*	M1D medium, 25 °C, 12 days, 110 rpm	Salicylic acid	Taxol	Anticancer activity	Subban et al. (2019)
24	*Dimorphosporicola tragani*	YES medium, 25 °C, 14 days, 110 rpm	5-azacytidine and Valproic acid	Dendrodolide E, Dendrodolide G, Dendrodolide I	Anticancer activity	González-Menéndez et al. (2019)
25	*Anteaglonium* sp.	PDB medium, 28 °C, 15 days, 160 rpm	Anacardic acid	Scorpinone, 1-hydroxydehydroherbarin, Ascochitine	Cytotoxic activity	Mafezoli et al. (2018)
26	*Drechslera* sp.	Liquid media, 25 °C, 12 days, 110 rpm	SAHA	Hexosyl-phytosphingosine and Benzophenone	Antifungal activity	Gupta et al. (2020)
27	*Diaporthe* sp.	PDB medium, 28 °C, 7 days, 150 rpm	Valproic acid	Xylarolide A and B, Diphoterine A	Anticancer activity	Sharma et al. (2018)
28	*Phoma* sp.	PDB medium, 28 °C, 14 days, 160 rpm	SAHA	(100 S)-Verruculide B, Vermistatin, and Dihydrovermistatin	PTPs inhibitors	Gubiani et al. (2017)
29	*Chaetomium* sp.	PDB medium, 28 °C, 14 days, 160 rpm	SAHA and 5-azacytidine	Isosulochrin	Anticancer and antimicrobial activity	Akone et al. (2016)
30	*Botryosphaeria rhodina*	PDB medium, 23 °C, 7 days, 160 rpm	5-azacytidine	Camptothecin	Anticancer activity	Vasanthakumari et al. (2015)
31	*Cophinforma mamane*	PDB medium, 27 °C, 14 days, 160 rpm	5-azacytidine, Sodium butyrate, Nicotinamide	3 thiodiketopiperazine	NA	Pacheco-Tapia et al. (2022)

Table 1 (continued)

Sr. No	Endophytic Fungi	Culture Conditions	Epigenetic Approach	Novel Bioactive Compounds	Biological Activity	References
32	*Macrophomina phaseolina*	PDB medium, 27 °C, 4 days, 150 rpm	OSMAC	Ergosterol peroxide, 1,4-benzene-diol, Butylated hydroxy toluene	NA	Singh et al. (2022)

affect secondary metabolite biosynthesis in fungi (Sun et al. 2019).

Mechanism of regulation of fungal secondary metabolites

Genome sequencing revealed two major outcomes while observing and analyzing these microbial genomes, especially those of fungi. First, there are some sorts of genetic clusters that are responsible for the production of secondary compounds, and these genes/clusters are majorly found in the sub telomere genomic region. Second, different types of enzymes that are encoded by these genes belong to various families, such as dimethylallyl tryptophan synthases (DMATSs), PKS, and non-ribosomal peptide synthetases (NRPSs) (Khaldi et al. 2010; Hautbergue et al. 2018). The production and release of these metabolites strongly depend upon the way host plants and fungi interact. Thus, in order to recognize these interactions between fungi and their host plant, the entire establishment and maintenance of the transcription factors supervising the BGCs expression in fungi is necessary (Mózsik et al. 2022). The secondary metabolite production process is highly complicated since it is synchronized with the morphology and phenotypic development of fungi. These metabolites are usually formed during the spore formation stage of development in fungi, subsequently after the adaptations to the environment, thereby entering the log phase of growth. Transcription factors control the transcription by interacting with the promoter sequence that interacts with different recognition sequences (target sequences), which are found in the promoter regions of various genes and control the transcription of SMs synthesizing genes. These factors may be broad domain factors that respond to certain signals, such as nitrogen and carbon supply, pH, or regulatory proteins, which have specificity for some BGCs. As a result, such regulatory networks synthesize SMs due to their response to normal cell metabolism and specific pathway inducers (García-Estrada et al. 2018). The bioactive metabolites with high potential are produced upon requirement only because of the very strict regulation of their biosynthesis. One notable example is the gene cluster ACE1, which is present in *Pyricularia oryzae*, a fungus causing rice blast disease, which is expressed for a brief period only when the fungus begins to penetrate the initial cell of a rice leaf (Collemare and Seidl 2019). In order to understand how two partners that are symbiotically associated interact with each other, the regulatory mechanism for genetic expression in endophytes and their respective hosts needs to be thoroughly analyzed. In symbiotic interaction, the genes that code for this might be controlled by certain regulatory factors, such as transcriptional factors (TFs), the clustering of genes, etc.

The research on the diverse genome and genetics of fungi has shown that the genes that code for biosynthesis pathways to synthesize SMs are grouped as clusters. Some genes that encode for certain factors or enzymes required for the pathway functions in the form of transporters, TFs, are found in these clusters too. According to the following observations, the structural organization of these fungal clusters can change depending on the environment. First, a few SMs that are extremely complex are synthesized when two or more clusters collaborate. Second, in a few cases, the clusters that are found close or adjacent to one another in the genome usually belong to different pathways. The chromosomal areas that are dynamic in nature or the telomeric regions typically contain these clusters. Although the mechanism by which these gene clusters encode the metabolites in unstable areas of DNA is yet unknown, a plausible theory is that they may be connected to the regulation of gene expression (Singh et al. 2024). Most of the TFs that are pathway-specific are found in these biosynthetic clusters, and they frequently control the expression and regulation of genes involved in their own biosynthetic pathways. Zn2Cys6-type TFs, which are widely distributed in the BGCs of fungi encoding for SMs, are the major pathway-specific TFs (Mózsik et al. 2022). Gene expression can be regulated by a variety of mechanisms in a global or specific pathway. The benefit of specifically regulating a gene's expression is that it helps in linking the gene clusters and the natural product stably as well as in the significant production of the desired SMs (for detection and structural elucidation). Given that single TFs regulate the entire genetic expression in a cluster collectively, overexpressing just one component will affect the transcription of the entire target cluster, simplifying laborious genetic manipulations (Khaldi et al. 2010). The gene clusters for SMs contain one or more TF-encoding genes as well as co-regulatory genes, which control the co-expression of the pathway-related genes locally. The expression of the genes for sterigmatocystin as well as aflatoxin in the *Aspergillus* species is regulated by AflR, which is the most studied transcription factor in an SM gene cluster (Bok and Keller 2004).

Two major categories of transcription factors—broad domain transcription factors (BDTFs) and narrow domain transcription factors (NDTFs)—further regulate the activities of the gene clusters in the secondary metabolism. In addition to acting on the gene clusters at genomic regions distinct from themselves, NDTFs can also act on the genes inside the cluster.AflR, a typical NDTF, which is basically a characteristic Zn2Cys6 TF, serves as an example of this. It regulates the gene clusters for sterigmatocystin and aflatoxin by binding to the 5′-TCG (N5) CGA-3′ palindromic sequence, an 11-bp motif found in the coding genes at the promoter regions in some *Aspergillus* sp. However, BDTFs are transcription factors that are global hierarchical control systems, which react to stimuli that have no direct relation with SM biochemical genetic clusters. Silent clusters are formed by stimulating the transcription process when there is no specific regulation. The non-clustered genes present in global regions encode certain global transcription factors, which help in the regulation of those genes that take no part in secondary metabolism. LaeA is the most widely used worldwide regulator of secondary metabolism in fungi (Hautbergue et al. 2018).

A methyl-transferase protein LaeA (loss of aflR expression), which was found in the nuclear region in the *Aspergillus* species, has revealed the regulation of secondary metabolite synthesis in filamentous fungi (Bok and Keller 2004). The LaeA protein is produced by a gene whose sequence analysis showed that it is related to arginine and histone methyl-transferases. The profiles depicting the transcription of the LaeA mutant, a wild strain and its complemented *A. fumigatus* strain were also compared in a study by Perrin et al. (2007). They demonstrated that out of the 22 species, almost 13 secondary metabolite gene clusters were under the transcriptional control of *laeA*. Seven among those *laeA*-controlled gene clusters were found within sub telomeric areas. *LaeA* might help in chromatin remodeling as is quite evident by the position of *laeA* regulating genes having sequences somewhat similar to methyl-transferases. *LaeA* is essential for fungal morphology, growth, and development in addition to the regulation of fungi's SM gene clusters. Various SM clusters can be targeted simultaneously, and this is made possible by globally manipulating the transcription factors (Perrin et al. 2007). The *laeA* was overexpressed in *Aspergillus* sp. to show this strategy, and it activated or boosted the production of numerous known bioactive metabolites (Pillay et al. 2022). The presence of other species, as well as changes in the supplies of nitrogen or carbon, pH, temperature, and light, all have an impact on SMs biosynthesis. In contrast to TFs that are specific to gene clusters, there are global transcription regulators that control the expression of a vast number of target genes spread throughout the genome, which are frequently used by fungi to respond to the above-mentioned stimuli. For instance, AreA influences the production of a wide spectrum of SMs, one of them being gibberellins, in addition to regulating vast sets of genes responsible for nitrogen catabolism corresponding to the quantity and quality of available nitrogen. Like this, PacC is a crucial regulator for regulating fungal pH in *Penicillium chrysogenum*, it promotes the biosynthesis of penicillin at an alkaline pH; in *A. nidulans*, it inhibits the expression of genes responsible for the synthesis of sterigmatocystin. The locus of aflR contains certain binding sites for CreA, which is a transcription factor that represses the catabolism of carbon and regulates the biosynthesis of aflatoxins (Collemare and Seidl 2019).

Regulation by nitrogen, carbon, pH, and light sources

A vital nutrient needed for growth and development is nitrogen. Microbes can metabolize a wide variety of nitrogen sources, enabling them to colonize different environmental niches and endure nutrient scarcity. Nitrogen metabolite repression, which results from the selective consumption of complex nitrogenous sources, activates the global regulatory circuit, activating transcription of genes involved in absorption and metabolism, and assures destruction (Gupta et al. 2017). Carbon is a crucial nutrient that has a significant impact on how microorganisms grow and evolve. Simple carbohydrates, including D-xylose, acetate, D-glucose, and sucrose, are preferred by fungi. In addition, when simple sugars were not available, other complex carbon sources, such as cellulose, fructose, pectin, xylan, glycerol, amino acids, and alcohol, were used. Ensuring the preferential use of glucose, cells cultured with simple sugars produce a carbon catabolism repression that tends to suppress the transcription of genes that catabolizes carbon (Adnan et al. 2017). Extracellular pH, a significant factor that modifies the degree of the expression of many different genes in fungi, has an impact on the proliferation of the cells. The gene regulatory system, regulated by a protein called PacC globally, modulates the expression of genes over different pH levels (Zhang et al. 2024a, b). PacC activation is regulated by the six pal gene products in alkaline environments (Orejas et al. 1995). For asexual spore formation in fungi, light is a crucial signal that induces conditions in the presence of red light while suppressing it at far-red light. The complex proteins that make up a complex regulatory network and have the LOV (light–oxygen–voltage) sensor domain, which detects blue light, and the PAS (Per-Arnt-Sim) domain, which is involved in protein dimerization, modify the way in which light signals are received and transmitted. Two important light-response regulatory proteins found in *Neurospora crassa*, White Collar-1 (WC-1) and White Collar-2 (WC-2), contain photoreceptors that dimerize when exposed to blue light and work as a transcription factor complex (white collar complex, WCC) (Beattie et al. 2018).

Cultivation-dependent approach for endophytic fungi

Endophytes can simultaneously help plants with nutrient uptake and utilization from the soil by biosimulating numerous substances and boosting their availability. These nutrients have been related to raising plant yields by promoting plant growth and development. Several endophytic fungi can also affect plant resistance to pathogenic organisms and insects and their defense mechanisms against abiotic stress factors, such as a lack of rain and excessive salt. Our understanding of the makeup of the microbiome and its interactions with host plants has significantly improved because of advances in next-generation sequencing and numerous omics approaches (Kimotho and Maina 2024). Approaches that depend on cultivation can be biotic or abiotic. A biotic strategy demands the employment of organisms to promote different kinds of natural products, whereas an abiotic approach may call for physical and chemical modifications. Optimizing the growth media nutritional composition, such as through its carbon source and amino acids, and improving its physiological conditions, such as its pH, temperature, and the concentrations of trace elements, are the critical determinants for the production of natural metabolites of a good quality and quantity (Pinedo-Rivilla et al. 2022).

Epigenetic activation, which requires adding substances that can regulate gene expression to the culture medium, is another approach for the cultivation-dependent technique. Recent studies showed that growing fungus in the presence of HDAC and DNMT compounds can activate cryptic BGCs, which can increase the chemical variety of natural products (NPs). 5-Azacytidine is the most well-known DNMT inhibitor for reactivating abnormally silenced genes through methylation by DNMTs and returning their normal activity (5-AZA). The suberoyl bis-hydroxamic acid (SBHA) and suberoylanilide hydroxamic acid (SAHA) are HDACs, which inhibit histone deacetylation and can promote gene transcription and expression in microorganisms (Qadri et al. 2017). The SMs are not present in the wild-type strain, but in the presence of epigenetic modifiers, there may be chances of production SMs. The gene expression of non-ribosomal peptide synthetase (NRPS) and PKS was demonstrated in a study by Qadri et al. (2017) on the endophyte *Muscodor yucatanensis* in the presence of the epigenetic modifiers SAHA and 5-azacytidine. The production of new compounds was observed, and a stable variation in the fungus for biological and morphological characters was also observed. The endophytic fungus *Nigrospora sphaerica* living in *Adiantum philippense* L when treated with epigenetic modifiers, such as 5-azacytidine, valproic acid, hydralazine hydrochloride, sodium butyrate, and quercetin, was found to create cryptic compounds with therapeutic effects (Xue et al. 2023). One of the most impressive advantages of using chemical epigenetic modifiers is that they do not require prior knowledge of specific genome features, unlike other genome-editing techniques, which require knowledge of the target sequence. Chemical epigenetic modifiers can be applied to large-scale, high-throughput screening operations to identify compounds that can modulate gene expression in specific ways (Pan et al. 2019).

OSMAC strategy to explore structural diversity of novel bioactive compounds

An alteration of the physiological growth conditions of a medium can lead to the production of a variety of compounds, which has medicinal importance for the endophytic fungi without the manipulation of the genetic makeup. The cultivation of a single fungi strain in optimized physiological factors, such as the addition of epigenetic modifiers or alteration of pH and temperature to produce different compounds from a single strain, gives the concept of OSMAC. Traditional endophyte screening frequently under-reveals the full metabolic capability of fungi and leads to the re-isolation of already well-known metabolites after endophytes have been cultured under normal laboratory conditions (Gao et al. 2020). The purpose of growing fungi for the OSMAC strategy is to activate the BGCs that are normally silent and not expressed during fermentation. The OSMAC method involves altering cultivation parameters, such as the medium composition (carbon/nitrogen ratio, salinity, metal ions) and physical parameters (pH, temperature, oxygen condition), or the addition of enzyme inhibitors/inducers, and biosynthetic precursors, in order to activate silent BGCs and broaden the metabolite pattern produced by endophytes (Fan et al. 2019).

The OSMAC method, also known as the systematic adjustment of culture conditions, has been successfully employed to access the range of SMs in fungi, leading to the identification of novel and bioactive metabolites (Gao et al. 2020). In another study, marine fungi were found to be present in all habitats and contribute significantly to marine environments. The number of secondary metabolite BGCs and the quantity of compounds isolated from microorganisms that have been chemically categorized varies significantly, according to later bioinformatic methodologies in genomics. Using an OSMAC method, eight different solid- and liquid-based culture mediums were used to cultivate the isolated epi- and endophytic fungi. Before being used in an untargeted LC–MS/MS-based molecular network employing automated (GNPS and ISDB-UNPD) and conventional manual dereplication techniques, the extracts were evaluated for their capacity to be cytotoxic and anticancer. According to the results of the UPLC–MS/MS, the chemical composition and bioactivity of the extracts significantly varied depending on the culture conditions. The bioactivity information made it possible to identify particular metabolites in extracts that were hazardous or anticancer (Wei et al. 2021). *Penicillium dangeardii*, an endophytic fungus, has 43 BGCs, found after a bioinformatic study of its genome, indicating the fungus's capacity to create a large number of SMs. However, this strain primarily generates significant yields of rubratoxins on their own under a variety of growth conditions, suggesting that the majority of the gene clusters are inactive. Epigenetic control and the OSMAC strategy were unsuccessful in revealing *P. dangeardii*'s cryptic gene clusters, most likely as a result of the organism's high rubratoxemia. Multiple genes encoding for additional PKSs were successfully triggered using a metabolic shunting technique that combined the deletion of the essential gene for rubratoxin production with a culture environment adjustment, enabling the detection of the trace substances. Thus, 23 novel compounds were identified, including siderophores, azaphilone monomers, dimers, and trimers with novel polycyclic bridging heterocycle and spiral structures. Some substances showed notable antioxidant, anti-inflammatory, or cytotoxic effects (El-Hawary et al. 2021). In another study by El-Hawaryet al. (2021) the OSMAC method was used to subculture an endophytic fungus called the *Aspergillus terreus* isolate that is connected to soybean roots in five different culture conditions. Utilizing molecular docking against the primary protease of COVID-19, the dereplicated metabolites were evaluated (Mpro). Metabolic profiling led to the identification of 18 molecules from distinct chemical classes. The most abundant classes were polyketides, isocoumarins, and quinones. Aspergillide B1 and 3a-Hydroxy-3,5-dihydromonacolin L have the greatest binding energy scores toward COVID-19 major protease (Mpro), and they interact strongly with Mpro's catalytic dyad, according to molecular docking studies (His41 and Cys145). The hunt for natural anti-COVID-19 compounds was made more effective by combining metabolomics and in silico methods. Aspergillide B1 and 3a-Hydroxy-3,5-dihydromonacolin L were identified as promising anti-COVID-19 medication candidates through a molecular docking investigation (Knowles et al. 2022).

Co-cultivation approach of endophytic fungi to explore structural diversity of novel bioactive compounds

Co-culturing limits several fungal species to a region with limited resources and territory, resulting in two significant interactions between them. First, they might biosynthesize compounds that might prevent the other fungi from growing (vice versa). Additionally, a fungus's pace of development can be adjusted, possibly accelerated, so that it consumes resources more quickly than its competing fungi, basically causing yet another type of stress. The true amount of SMs, however, might go unnoticed if the silent BGCs are not stimulated, as this would make them active. Co-culturing is important because it more accurately resembles the fungi's natural environment, driving them to compete for survival and possibly revealing novel bioactive substances that are not synthesized under typical monoculture environments (Tiwari and Bae 2022). The co-cultivation of various strains of endophytic fungi is a promising strategy to activate silent gene clusters to express their genes. For example, when

Podocarpus gracilior leaves were co-cultivated with *A. terreus*, the gene expression for taxol synthesis was restored (Ding et al. 2018).

In another study, Ding et al. (2018) isolated three endophytic fungi from *Rumex gmelini* Turcz (RGT), which are the *Aspergillus* species, *Fusarium* species, and *Ramularia* species, and they produced secondary bioactive compounds from their hosts. However, these fungi's capacity to manufacture active ingredients considerably declined and was challenging to restore after being cultured alone for a long time. Co-culturing RGT-tissue culture seedlings with their fungus endophyte resulted in the production of more bioactive SMs. With regard to the accumulation of bioactive components in RGT seedlings, the *Aspergillus* sp. was the most influential fungus. The yields of chrysophaein, resveratrol, chrysophanol, emodin, and physcion were 3.52, 3.70, 3.60, and 1×10^4 mL^{-1}, respectively, after co-culturing *Aspergillus* sp. spores for 12 days with RGT seedlings that had taken root for 20 days. The highest musizin yield, 0.289 mg, was not present in the control groups. Based on the findings, co-cultivating RGT seedlings with endophytic fungi could significantly improve the synthesis of beneficial SMs (Olano et al. 2008).

Ribosomal engineering approach of endophytic fungi to explore structural diversity of novel bioactive compounds

Only 150 of the over 23,000 SMs produced physiologically by microbes are employed in the pharmaceutical, agricultural, or other industries. The production of pharmaceuticals for commercial use involves chemical synthesis or semi-industrial procedures. These microbial strains require advanced production techniques to ferment the chemicals and separate the components. These conditions result in high production costs for the isolation and extraction of these substances, which limits the extraction of SMs from naturally occurring fungi. It unequivocally shows that the cost cannot be decreased without increasing production and that the cost rises when a new chemical is processed. *P. chrysogenum* now produces 1000 times more penicillin than 70 g/L, up from 60 mg/L previously. Using secondary metabolite genetic engineering, the amount of a particular chemical or a collection of target compounds can be changed. Furthermore, there are two main strategies that newly discovered chemicals use to improve their production. At first, one or more genes were employed to get around the precise steps that were slowing down the process, obstructing rival pathways, and lowering catabolism. The second strategy tries to regulate the numerous biosynthetic genes in order to change how regulatory genes are expressed (Pickens et al. 2011; Rastegari et al. 2019; Qin et al. 2022). The terpenoid indole alkaloid pathway is the next area of focus for gene-splicing studies since terpenoid and terpenoid indole alkaloids, including the antitumor alkaloids vinblastine, vincristine, and camptothecin, have commercial value (Satish et al. 2020). Isoquinoline alkaloids, a well-known secondary plant metabolite that includes important drugs, such as morphine and codeine, serve as a replacement for the essential drug class (Wang et al. 2022). The function of 2-C-methyl-D-erythritol-4-phosphate (MEP) in plastidial terpenoid biosynthesis, including the generation of monoterpenes and diterpenes as well as red fat-soluble pigments, has recently come to light, leading to the duplication of numerous genes in this pathway (Li et al. 2022).

Genetic engineering can be applied to improve the production of SMs by endophytes and the synthesis of SMs of good quality and quantity. In addition, the development of modern technologies, such as DNA sequencing, transcription profiling, metabolomics, genomics, proteomics, and metabolite profiling, has opened up new possibilities for engineering microbes for the synthesis of natural chemicals in high yields (Kaur et al. 2024). It is well-known that the synthesis of CoASH (Coenzyme A) active glyceride precursors promotes the conversion of some SMs into beneficial products, including erythrocin, macrolides produced by *Streptomyces*, polyether antibiotics isolated from *Streptomyces cinnamonensis*, and benzoisochromane polyketide antibiotics (Torres-Haro et al. 2021). Proteomics, random mutation, and metabolic engineering are all effective methods for increasing the yield of SMs. The most common genetic method used today to increase the production of bioactive substances is metabolic engineering. Using recombinant DNA technology, this technique modifies chemical expression (and in some cases, overexpression) and production (Abbas et al. 2024). Haleem et al. found that *Saccharomyces cerevisiae* may be used to introduce the genes from *Ralstonia eutropha* that code for the synthesis of polyhydroxybutyrate (PHB), a polymer utilized in the production of biodegradable plastic. The resulting recombinant yeast produced a polymer with a 0.44% of cell dry weight, a substantial improvement above the wild strain's production of a polymer with a 0.08% cell dry weight (Sezonov et al. 1997). Genes that are complex in the production of SMs are typically targeted for removal or inactivation in order to provide novel substrates. Additionally, these mutations may serve as hosts for the expression of heterologous genes that may aid in boosting the creation of novel substrates. *Streptomyces pristinaespiralis* produces the streptogramin antibiotic pristinamycin II by combining two PIIA and PIIB substrates in an 80:20 ratio. During the conversion of the PIIB molecule into PIIA, the *snaA* and *snaB* genes must be inserted as an additional copy to the chromosome to transform heterodimeric monooxygenase, which undoes this mutation. The *snaA/snaB* gene increases the levels of PIIA and pristinamycin while not increasing the overall quantity of pristinamycin, which improves the ability to generate this substrate (Uzma et al. 2018). It is possible to direct the production of SMs to rise from other primary metabolites, such as G3P, lysine, or 2-exoglutarate, using structural biosynthesis genes. Due to excessive clavaminate synthase gene cas2 expression or chromosomal integration, *S. clavuligerus* produced five times as much clavulanic acid. An additional enhancement of up to 23-fold was produced by the simultaneous integration of the clavaminate synthase gene cas2 and the activated crossover chromosome ccaR (Pickens et al. 2011).

To highlight the relationships between the genome and metabolome, various studies have used transcriptomes as a middle step in addition to the direct observation of the fungal secondary metabolite or genetic creation of cryptic clusters. The methodologies used in this case have a dual focus on microarray and RNA-seq (or whole-transcriptome shotgun sequencing). These methods are useful for locating gene clusters that are co-regulated or clusters that are separately involved in the production of several SMs but whose expressions are linked. Transcriptomics can be modified to specifically find gene clusters that are distributed across distinct chromosomes. The most common approaches are genome-based techniques and metabolomics, which are occasionally combined with transcriptome studies. However, proteomic analysis can also be used to study the secondary metabolomes of fungi. Proteomic methods are useful to account for the potential post-transcriptional modifications that might occur during the synthesis of fungal SMs. 2D PAGE–LC–MS/MS analysis can be used to detect proteins. An enhanced method, called Proteomic Investigation of Secondary Metabolism (PrISM), selects the high-molecular-weight proteins after 2D SDS-PAGE separation to target the primary enzymes of the secondary metabolome. Following the characterization of the enzymes, the pertinent gene clusters are identified, the SMs that correlate to those enzymes are characterized, and their PCR (polymerase chain reaction) primers are made (Hautbergue et al. 2018).

Activator or precursor feeding approach of endophytic fungi to explore novel bioactive compounds

Since more than 30 years ago, a number of fundamental biotechnological techniques have been created, including the cell culture, callus culture, organ culture, and micropropagation of whole plants. These techniques are an alluring alternative to create economically significant and valuable SMs from field-grown plant materials. Their use in the large-scale synthesis of the desired SMs has so far had very little commercial success because of the culture's low productivity (Loyola-Vargas and Ochoa-Alejo 2024). The efficacy of a variety of techniques, including hairy root culture, metabolic engineering, plant cell immobilization, bio-transformation, precursor feeding, elicitation, large-scale cultivation in bioreactor systems, screening and selection of high-yielding lines, optimization of culture media composition and physical parameters, etc., has been investigated. To induce novel SMs or enhanced production as well as secondary metabolite accumulation in in vitro plant tissue cultures, elicitation is one of the most well liked and effective biotechnological techniques. Elicitors are biotic or abiotic compounds that, although coming from various chemical families and having no shared chemical structure, can stimulate or enhance the production of a certain secondary metabolite. Elicitors act as signals, and elicitation starts with the plant cell membrane's elicitor-specific receptors detecting the signal. The start of a signal transduction cascade that alters the level of expression of several regulatory transcription factors and genes as well as rate-limiting genes of the secondary metabolic pathways results in an increase in the synthesis and accumulation of SMs. Undifferentiated cells are exposed to various elicitors, which increases the production of SMs, but this method has some disadvantages, such as low productivity when compared to organ cultures, genetic and biosynthetic instability in long-term cultures, irregular response to the same elicitor, etc. (Gorelick and Bernstein 2014). Elicitors are a group of biotic or abiotic signals with a variety of chemical compositions that, when given in tiny doses to a living cell system, encourage or expedite the creation of a specific secondary metabolite by eliciting defense- or stress-induced responses. Elicitors can be categorized as either abiotic or biotic based on their "nature" or according to their "origin," such as external and endogenous elicitors. Biotic elicitors are either unprocessed extracts or products that have undergone some purification and are derived from either pathogens (fungi, bacteria, or yeast) or the plant itself. They either have a particular composition, such as polysaccharides, glycoproteins, inert enzymes, pure chitosan (CHI), pectin, chitin, alginate, curdlan, xanthan, and elicitin, or a complex composition, such as yeast extract (YE) and fungal extract (Halder et al. 2019). Light, UV rays, heavy metal salts ($Ag_2S_2O_3$, $AgNO_3$, $CdCl_2$, $CuCl_2$, $CuSO_4$, $VOSO_4$, $NiSO_4$, selenium), temperature fluctuations, osmotic stress brought on by mannitol, sorbitol, sodium chloride, potassium chloride, cadmium chloride, polyvinyl pyrrolidone (PVP), etc., are examples of abiotic elicitors. Intracellular signaling molecules, including systemin, jasmonic acid (JA), methyl jasmonate (MJ), salicylic acid (SA), acetyl salicylic acid (ASA), and salicylic acid, are also used as elicitors. In hairy root cultures of *Datura metel*, *Staphylococcus aureus* and *Bacillus cereus* were utilized as biotic elicitors, which increased root biomass accumulation but decreased atropine accumulation. In *S. lateriflora* hairy roots, wogonin accumulation was nearly tripled after *Pectobacterium carotovorum* lysate was added during the stationary phase of the culture (30 mg g^1DW) (Scherlach and Hertweck 2009).

Epigenetic modulation approach to explore novel bioactive compounds

When the genome is altered, the proteome and transcriptome are also altered, and the cryptic gene(s) product is activated. The alteration may occur at the genomic level and include gene knockout, promoter exchange, and epigenetic modulation (methylation, acetylation, and phosphorylation (Cheynier et al. 2013). Modifications at the transcriptome level include changes to metabolomics and the regulation of transcription factors (either through overexpression or repression). By activating recently identified genes or by suppressing previously active genes, epigenetic modulators may change the expression of gene(s) involved in the manufacture of natural metabolites (Harrison and Dexter 2013; Schüller et al. 2022). Nicotinamide controls epigenetic processes by inhibiting the NAD^+-dependent enzyme HDAC. Nicotinamide treatment of the endophytic fungus *Eupenicillium* sp. LG41, isolated from *Xanthium sibricum*, resulted in the activation of a new gene and the production of the decalin-containing substances eupenicinicol C and D in addition to the previously known substances eujavanicol A and eupenicinicol A (Li et al. 2017; Jasim et al. 2019).

The endophytic *Diaporthe* sp. PF20 from *Piper nigrum* L. was discovered to produce piperine using liquid chromatography. The isolate PF20 was additionally subjected to epigenetic therapies along with previously characterized piperine-producing *Colletotrichum* sp. and *Mycosphaerella* sp. from the same plant. Intriguingly, the HDAC inhibitor suberohydroxamic acid was used to boost the synthesis of piperine in PF20. Here, the use of epigenetic modulators to increase the capacity of endophytic fungi to synthesize phytochemicals is a novel approach. As a result, there are creative approaches to maintain the biosynthetic competency of endophytic fungi, which is a challenging task. Although the production of piperine by endophytic fungi has already been confirmed, the multiplexing of this feature by an epigenetic modulator as demonstrated in PF20 is highly intriguing. This is due to the fact that only this particular organism among the selected isolates was demonstrated to respond to piperine enrichment through epigenetic alteration (Toghueo et al. 2020). Chemicals that can inhibit the activity of enzymes, such as histone deacetylase, DNA methyl-transferase, and proteasome, have been found to be effective in promoting the production of specialized metabolites in fungi. In addition, substances, such as organic solvents, metals, and plant extracts, have also been found to have the ability to alter the way fungi produce these metabolites. These changes in fungal metabolism can result in the creation of new compounds or increased production of existing ones, which has many practical applications in various industries (Chiang et al. 2009).

Genomic mining approaches

The combination of proteomic and genomic approaches emerged as an effective tool for discovering new natural products. In silico prediction can reveal the chemical structures encoded by orphan genetic loci. These tools have proven to be extremely useful for analyzing the vast amount of available genomic information. They automatically analyze genetic data, which serves as a starting point for further experimental investigations (Brakhage and Schroeckh 2011; Scherlach and Hertweck 2021). For example, software tools like rBAN30 can simulate the retro-biosynthesis of non-ribosomal peptides (NRPs) from their chemical structure, predicting the necessary enzymatic machinery and prioritizing promising BGCs for novel compound discovery (Ming et al. 2023). Such tools have become an invaluable resource for researchers, enabling them to discover new natural products. A systematic analysis of genome mining for BGCs typically involves the following steps:

Genome sequencing and annotation

The first step in genome mining is to obtain the genomic sequence of the endophyte and to annotate it to identify genes and their functions. High-throughput sequencing technologies, such as Illumina or PacBio, are used to sequence the fungal genome. For the prediction of coding genes in a genome, software such as Maker2 (v2.31.9) is used. tRNA genes can be predicted using tRNA-scan-SE software (v2.0), while rRNA genes can be predicted using Barrnap software (v0.8). Once the coding genes have been predicted, they can be annotated using various databases, such as NR, Pfam, Swiss-Prot, COG, GO, and KEGG (Chavali and Ree 2018).

Identification of BGCs

Once the genome has been annotated, various bioinformatics tools can be used to identify potential BGCs. These tools typically search for characteristic gene clusters that are involved in the synthesis of a particular class of natural products. There are several bioinformatics tools and algorithms available for the identification of BGCs in microbial genomes. For instance, Antibiotics and Secondary Metabolite Analysis Shell (AntiSMASH) is used for the automated identification and annotation of BGCs in microbial genomes. It uses a combination of algorithms and databases to predict the presence and function of biosynthetic genes and to annotate the predicted BGCs with information on the biosynthetic pathway, gene clusters, and domain structures (Schüller et al. 2022).

Verification of BGCs

Once potential BGCs have been identified, they must be verified experimentally. The first step would be to conduct gene expression analysis through RNA sequencing (RNA-seq) analysis, quantitative real-time PCR (qPCR), or other gene expression profiling methods. If the genes are expressed under the appropriate conditions, this is a good indication that the BGC is active. The next step would be chemical analysis, which involves analyzing the chemical products produced by the BGC. This can be achieved through various methods, such as mass spectrometry or nuclear magnetic resonance (NMR) spectroscopy. If the expected compound(s) are detected in the endophyte, this provides evidence that the BGC is functional (Ning et al. 2022). For further verification and confirmation, gene knockout or knockdown strategies can be applied to disrupt the BGC. This can be performed through genetic engineering methods, such as CRISPR/Cas9 or RNA interference (RNAi). If the knockout or knockdown of a specific gene results in the loss of the expected metabolites, this provides strong evidence (Kjærbølling et al. 2019). Another approach is to conduct complementation experiments to verify the function of a BGC. This involves introducing a functional copy of a gene or genes into a strain that lacks the BGC, to see if it can produce the expected compound(s). Once a BGC has been verified, it can be further characterized to understand the specific biosynthetic pathway involved in the production of the natural product. This can involve identifying the individual enzymes involved in the pathway as well as their substrate specificities and reaction mechanisms (Abo-Kadoum et al. 2022; Wojnowska et al. 2023).

Optimization of condition for expression of BGCs

Once the biosynthetic pathway has been characterized, it can be optimized for the improved production of the natural product. This can involve the genetic engineering of the host organism as well as the optimization of the culture conditions and fermentation process (Bhaskar et al. 2024; Klumbys et al. 2024).

Conclusions

In order to meet the growing demand for novel potent bioactive molecules, it is now more important than ever to look into alternate sources of epigenetic modulators and innovative approaches to high-throughput drug discovery. The bioprospection of endophytic fungi for such purposes seems the best selection since endophytic fungi are producers of various well-known categories of drugs, such as antibiotics, anticancer, antimicrobial, and antifungal drugs, and biofuel-related hydrocarbons. It has been reported that endophytic fungi's epigenetic alterations are excellent strategies for accelerating the synthesis of cryptic bioactive molecules. In fact, using small molecule elicitation, more than a thousand novel specialized molecules have been discovered to date. With the help of technology, it is also possible to improve the yields at which a number of significant and rare specialized compounds are produced. In addition to these benefits, the use of small molecule elicitors is an easy, affordable, and widely applicable technique helpful for both the biotech industry and laboratories with limited resources. Genomic mining approaches are a useful tool for discovering new natural products. The process typically involves genome sequencing and annotation, the identification of BGCs, the verification of BGCs through gene expression analysis and chemical analysis, the optimization of conditions for BGC expression, and the further characterization of the biosynthetic pathway. It is noteworthy that fungi can produce a variety of specialized metabolites under various environmental conditions, the number of which is only constrained by our imagination.

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Declarations

Competing interests The authors declare no conflicts of interest.

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